

CSC6124 — Graph Computing

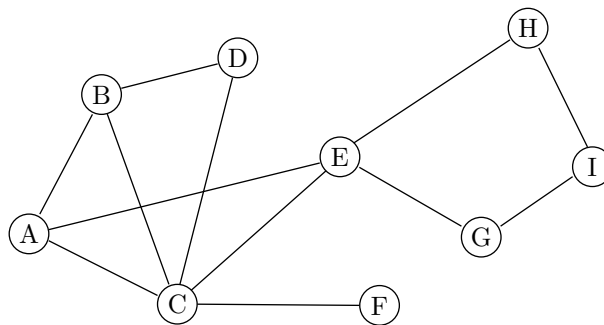
Quick Exercise: Page 32

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Exercise

Source: Topic 1: Graph Traversal-1, page 32 (Quick exercise).

Can you show the result of iterative depth-first traversal?



Given pseudocode (lecture note, page 32):

```
void Graph::depth_first_traversal( Vertex first ) {
    bool<Vertex> visited(|V|, false);
    visited[first] = true;
    stack<Vertex> stack;
    stack.push( first );

    while ( !stack.empty() ) {
        Vertex v = stack.top();
        stack.pop();
        print the vertex v;
        for ( Vertex w : v.adjacent_vertices() ) {
            if ( ! visited[w] ) {
                visited[w] = true;
                stack.push( w );
            }
        }
    }
}
```

Answer

[Write your independently completed answer here.]

Working (if required)

[Insert your own working or explanation here, if required.]